

Dynamic Asset Allocation Competition

Assessment of the current strategy and a robust gameplan for maximising the composite score

Overall assessment. The proposed architecture is sensible and likely competitive. Its strongest features are cross-sectional signal construction, volatility targeting, adaptive signal confidence, and drift-aware trading. Its main weaknesses are horizon handling, noisy IC adaptation, correlation blindness, and treating execution constraints after rather than inside portfolio construction.

Preferred gameplan

1. Replicate the scoring rule exactly. Optimise the full competition score rather than any single metric. Sharpe and volatility-target accuracy contribute 54 of 82 points, so they should dominate model selection.
2. Identify each signal's true horizon. Estimate cross-sectional information coefficients for each future day k , not only cumulative k -day returns. Overlapping 10-day returns create serial dependence and can materially overstate ordinary t -statistics.
3. Build separate signal sleeves. Signal 1 should probably use a slower decay or ten staggered sub-portfolios, while Signal 2 should trade faster. A single five-day smoothing rule and fixed 0.75/0.25 blend do not reflect their different horizons.
4. Construct weights using covariance. Inverse-volatility weighting ignores correlations. Use a shrinkage covariance matrix and optimise expected return, risk, turnover and execution limits jointly.
5. Use robust signal transformations. Cross-sectional ranks or winsorised z -scores are usually safer than raw rolling time-series standardisation. Cross-sectional demeaning is appropriate for creating a market-neutral signal sleeve.
6. Target volatility conservatively. Estimate individual volatilities relatively quickly, correlations more slowly, smooth leverage changes, and include realised-volatility feedback. Unlimited leverage should not be treated as economically unlimited because estimation error and tail losses remain binding.

$$IC_k = \text{corr}_i(s_{i,t}, r_{i,t+k})$$

$$w_t = \arg \max_w [\mu_t^T w - \lambda w^T \Sigma_t w - c \|w - w_t^{\text{drift}}\|_1]$$
$$\text{subject to } |w_i - w_{i,t}^{\text{drift}}| \leq 0.05$$

Assessment of the current strategy

Strong components

- Cross-sectional demeaning of the signal scores.
- Signal clipping and smoothing to reduce outlier sensitivity and turnover.
- A drift-aware no-trade region to avoid paying costs for negligible reallocations.
- Dynamic volatility targeting rather than fixed gross exposure.
- Separate confidence estimates for the two signals in the newer implementation.

Main concerns

- The low-risk sleeve is not automatically valuable. Once it is leveraged to the same 2% volatility target, only its Sharpe, diversification value and tail behaviour matter. It can dilute a stronger

signal sleeve.

- The 252-day rolling IC t-stat may be much noisier than it appears. If Signal 1 predicts ten-day returns, the window contains only about 25 non-overlapping ten-day observations. Overlapping observations inflate apparent confidence unless corrected.
- Automatic signal reversal is dangerous. A negative rolling t-stat can arise from noise. Reversal should require strong and persistent evidence, with shrinkage and a dead zone around zero.
- Continuous arctangent scaling may create unnecessary turnover. Small changes in estimated confidence should not always trigger portfolio changes.
- The 0.05 trading constraint should be embedded in optimisation. Computing unconstrained weights and clipping afterward changes risk, neutrality and signal exposure unpredictably.
- EWMA half-life 10 is plausible for variances but unstable for correlations. Estimating a 20-asset covariance matrix this quickly can generate noisy weights and excessive leverage.

Highest-priority changes

Priority	Change	Reason
1	Audit the IC calculation and t-statistics.	A mistaken horizon or overstated significance contaminates the entire strategy.
2	Separate the two signal holding horizons.	Signal 1 and Signal 2 should not be forced through the same smoothing and turnover profile.
3	Replace inverse volatility with covariance-aware optimisation.	Risk depends on correlations, not only individual asset volatility.
4	Optimise with costs and the 0.05 trade cap directly.	Post-hoc clipping produces a portfolio different from the intended one.
5	Validate robustness through walk-forward testing.	Test reversed signals, doubled costs, correlation spikes and volatility-estimation errors.

Bottom line. The existing design is a good baseline, but the likely edge will come from statistical discipline rather than adding more components: correct horizon estimation, conservative confidence shrinkage, covariance-aware allocation, and execution-aware optimisation. Those changes directly target the parts of the score with the largest weights while reducing hidden-period failure risk.